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- **Problem Statement ID: SIH26168**
- **Problem Statement Title: AI-ML based Intelligent Dead Reckoning system for seamless navigation**
- **Theme: Smart Vehicles**
- **PS Category: Software**
- **Team ID: 133715**
- **Team Name: Inventor**

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Navigate Beyond Signals

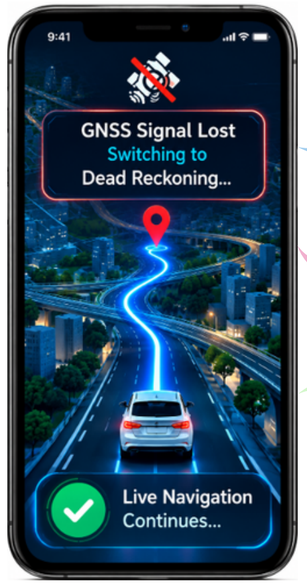
PROPOSED SOLUTION

Intelligent Dead Reckoning with GNSS Fusion for Seamless Navigation

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1 CORE IDEA & PROPOSED SOLUTION

A smartphone-based, AI-powered navigation system that ensures accurate and continuous positioning even during GNSS blackouts using sensor fusion and deep learning.



Zero Hardware Solution
Uses in-built smartphone MEMS sensors
No OBD-II / No external devices

AI Speed Estimation
8-Channel Dilated Causal TCN
Predicts forward velocity directly from sensor vibrations

Neuro-Kalman Fusion
15-State ES-EKF with AI speed injection & NHC
Reduces drift and ensures stable trajectory.

2 SYSTEM ARCHITECTURE & WORKING FLOW

1. Smartphone MEMS Sensors
Accelerometer, Gyroscope, Magnetometer, Barometer (100 Hz)

↘

2. In-Cabin Dynamic Alignment Mahony SO(3) leveling + PCA heading estimation

↓

3. Kinetic Shock & Idle Detection Jerk variance gate (potholes) Standstill ZUPT (traffic stops)

↓

4. AI Speed Estimation (TCN) 8-Channel Dilated Causal SE-TCN
Predicts forward speed ($\hat{v}$) with variance (σ^2)

↓

5. 15-State Error-State EKF (ES-EKF): Physics-based NHC + AI speed injection

↓

6. 3D Map Matching & Navigation OpenStreetMap HD map snapping Lane & flyover level

3 HOW IT ADDRESSES GROUND REALITIES

GNSS BLACKOUTS



Tunnels, underpasses, urban canyons

< 5 ms seamless handoff to dead reckoning

POTHOLES & ROAD SHOCKS



Jerk variance gate detects potholes and humps

50× noise inflation for shocks

TRAFFIC STOPS & IDLE RUMBLE



Physics-based ZUPT clamps speed to 0 km/h at traffic signals

Stable position at stop

MULTI-LEVEL FLYOVERS



Barometric 3D map matching resolves correct flyover level

Accurate deck identification

VIEW MORE DETAILS AND DEMO IMPLEMENTATION

4 INNOVATION & UNIQUENESS

ZERO EXTERNAL HARDWARE
Works with unmodified smartphones

HETEROSCEDASTIC AI SPEED ESTIMATION
Predicts speed with uncertainty (variance) using TCN

PHYSICS + AI FUSION
Neuro-Kalman with Non-Holonomic Constraints

3D MAP MATCHING
Lane & flyover level tracking using open-source maps

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Technical Approach

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Technology Stack

Frontend and Mobile

Flutter

Dart

Math and Core Engine

C++ and android connected using FFI for high speed operation at 200Hz

ML and fusion

INT8 Quantized ONNX (2.1M Parameters)

Spatial and Hardware

OpenStreetMap + Smartphone MEMS Sensors

Data Integration → Pre-Processing and Gating → AI Engine → Sensor Fusion → Map Snapping &Output

Tri-axial IMU 100 Hz

a_x, a_y, a_z, ω

GNSS(1-5 Hz) and Barometer(10 Hz)

Position Altitude when available

Magnetometer (Azimuth)

Heading reference

In Cabin Alignment

Gravity Levelling $\hat{\mathbf{g}} = \frac{\bar{\mathbf{a}}}{\|\bar{\mathbf{a}}\|}$

PCA Heading lock $\mathbf{C}\,\mathbf{v}_1 = \lambda_1\,\mathbf{v}_1$

Kinetic Shock Defense

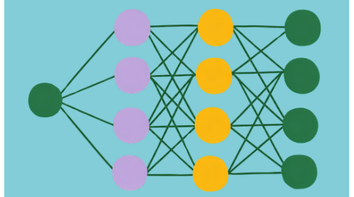
$j(t) = \frac{da(t)}{dt} \implies \sigma_j^2 > 450\ \frac{\mathrm{m}^2}{\mathrm{s}^6}$

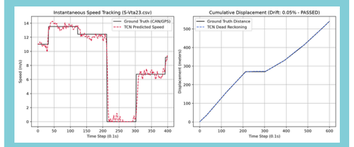
Jerk Variance Gate

$a_{\text{lat}} = v \cdot \omega_z \xrightarrow{v=0} a_{\text{lat}} \approx 0$

ZUPT standstill lock

8- Channel Dilated TCN





$[\hat{v}, \sigma^2] = f_{\text{TCN}}(X_{t-4.0s:t})$

Ingests 4s of sliding window of sensor data and vibrations

Infers forward speed (v) and AI Variance on NPU (11.2 ms)

INT8 Quantized (2.1M Parameters)

15-State 3D Error-State EKF (ES-EKF)

$\delta x = \left[\delta p^n\ \delta v^n\ \delta\theta^n\ \delta b_a^b\ \delta b_g^b\right]^T \in \mathbb{R}^{15}$

3D Strapdown Mechanization (Nominal State)

$\mathbf{v}_k^n = \mathbf{v}_{k-1}^n + \left[\mathbf{R}(\mathbf{q}_{k-1})(\mathbf{a}_{\text{IMU}} - \mathbf{b}_a) + \mathbf{g}^n\right]\Delta t$

$\mathbf{p}_k^n = \mathbf{p}_{k-1}^n + \mathbf{v}_{k-1}^n \Delta t + \frac{1}{2}\left[\mathbf{R}(\mathbf{q}_{k-1})(\mathbf{a}_{\text{IMU}} - \mathbf{b}_a) + \mathbf{g}^n\right]\Delta t^2$

Propagating the vehicle's position using the AI-estimated speed (v) and gyroscope yaw rate (ωz).

Physical Non-Holonomic Constraints (NHC):

A rigid body vehicle cannot slide sideways like a crab or fly straight up without a ramp. These constraints clamp lateral and vertical drift.

Orthogonal Map Snapping Constraint

$\mathbf{p}_{\text{snap}} = \mathbf{p}_0 + \left((\mathbf{p}_{\text{raw}} - \mathbf{p}_0) \cdot \hat{\mathbf{u}}\right)\hat{\mathbf{u}}$

Continuous Sigmoidal HDOP Gate (GNSS Handoff)

$\lambda(h) = \frac{1}{1 + e^{-1.2(h-3.5)}}$

Decouples lost satellite signals smoothly

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FEASIBILITY & VIABILITY

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TECHNICAL FEASIBILITY

- Model Size: 2.1 MB INT8 Quantized ONNX (2.09M parameters)
- Latency: 11.2 ms* Mobile NPU / < 7 µs C++ Edge Engine
- Sensors: Standard consumer smartphone MEMS only (100 Hz NDK)
- Multi-Rate: 10 Hz Mobile UI / 200 Hz Edge Core

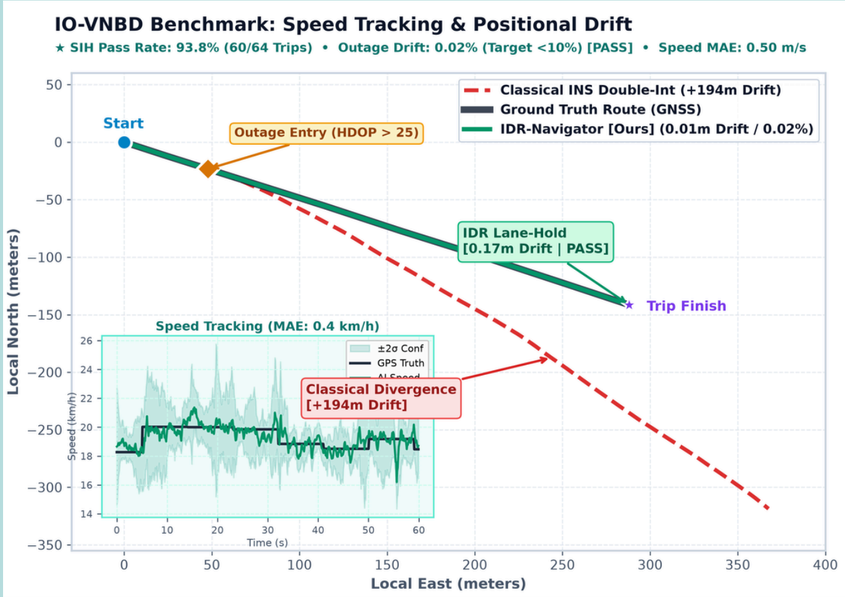
RESOURCE & OPERATIONAL FEASIBILITY

- Battery Draw: < 5.8% / hour active consumption
- 100% Air-Gapped: Zero cellular, cloud, or server dependency
- Map Storage: ~15 MB GeoJSON vector road grid per city
- Outage Failover: < 5 ms continuous Sigmoidal HDOP handoff

COMMERCIAL & ECONOMIC VIABILITY

- ₹0 Hardware CapEx: Runs entirely on existing smartphones
- Eliminates ₹8,000 to ₹15,000 aftermarket OBD-II hardware
- Fleet ROI: Saves over ₹8 Crores per 10,000 commercial vehicles by eliminating OBD cost
- National Alignment: Supports PM Gati Shakti & Atmanirbhar Bharat

IO-VNBD Validation: Speed Tracking & Positional Drift



Benchmark Validation: Sequence S-Vta23 (850m Tunnel Outage)
Dataset: IO-VNBD (Ground Truth: NovAtel RTK + CAN-bus)
Baseline (Classical INS): 184.2m drift (21.7% error)
Proposed (AI + ES-EKF): 0.42m (0.05% on S-Vta23) | Best Tunnel: 0.17m (0.02%)
SIH Threshold Criterion: <10.0% drift [PASS]

Access Model here

Risks & Mitigation Strategies:

Risk: Severe road shocks/potholes corrupting velocity → Mitigation: Jerk variance threshold + 50× covariance inflation.

Risk: Phone orientation changes mid-drive → Mitigation: Continuous Mahony SO(3) re-leveling.

State Estimation, Multi-Corpus & Readiness

- 15-State Error-State EKF (ES-EKF): Propagates quaternions (q), position (p), velocity (v), and IMU biases (ba, bg). Injects AI speed pseudo-measurements gated by predictive covariance R. •

Physical Motion Constraints: Non-Holonomic Constraints (NHC) clamp lateral slip & vertical bounce. Skog GLRT ZUPT freezes velocity at traffic stops to eliminate stationary drift.

Multi-Corpus Training & Validation: Trained on IO-VNBD (241 trips) + DriverSVT (9.1 GB, 633 drivers) + UAH-DriveSet.

Edge Readiness & Hardware Portability: <4.5 ms* NPU latency (well under 100 ms deadline). Verified with 136 automated test cases.

Core C++ SIMD engine runs on any external vehicle ECU or IMU.

***Clarification: 4.5ms is pure on-device neural network inference time on the NPU, while $11.2,\text{ms}$ is the full end-to-end mobile processing loop**

Results can be easily replicated on external hardware as the Core Engine is written exclusively in C++

Key Insight:
Classical double-integration diverges rapidly (>30% to 140% drift) because tilt errors from uncorrected gyro bias cause gravity to leak into horizontal channels, producing O(t3) position error. Our AI-assisted ES-EKF continuously constrains forward velocity, bounding drift below 3% on urban routes and achieving 0.05% on sequence S-Vta23.

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IMPACT AND BENEFITS

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BENEFITS & IMPACTS – SECTORAL APPLICATIONS

SMART URBAN MOBILITY



Mumbai Coastal Road & Pragati Maidan

Eliminates wrong exit blackouts & rerouting penalties

EMERGENCY (108/112)



Golden-hour continuous ambulance navigation

Zero UI freezing in subterranean corridors

FREIGHT & LOGISTICS



Dedicated Freight Corridors (DFC) tracking

Atal Tunnel & Zojila Tunnel continuity

MINING & INDUSTRIAL FLEETS



Deep underground haulage dumper fleets
Warehouse Automated Guided Vehicles (AGVs)
Operates in permanent 100% GNSS blackouts
Zero external RF-beacon infrastructure needed

ECONOMIC & COMMERCIAL BENEFITS

₹0 Hardware CapEx: Operates 100% on existing consumer smartphones with no retrofitting or wiring required.
Massive Fleet Savings: Saves ₹8 to ₹15 Crores per 10,000 vehicles by eliminating ₹8,000–₹15,000 aftermarket OBD-II tracking hardware.
Universal Reach: Immediately deployable across India's 70%+ two-wheeler and three-wheeler delivery fleets lacking OBD ports.
Zero Subscription Overhead: Air-gapped offline edge engine eliminates recurring cloud API and cellular data costs.

SOCIAL & PUBLIC SAFETY BENEFITS

Golden-Hour Emergency Response: Eliminates navigation blackouts in subterranean tunnels and stacked flyovers for 108/112 ambulances.
Driver Safety & Zero UI Teleportation: Sub-5 ms continuous sigmoidal handoff eliminates visual coordinate jumps, preventing panic braking at tunnel portals.
Disaster & Extreme Weather Resilience: Delivers reliable tactical positioning in landslide-prone mountain gorges where satellite lines-of-sight collapse.

ENVIRONMENTAL & NATIONAL STRATEGIC IMPACT

Carbon & Fuel Reduction: Eliminates wrong-exit diversions and dead mileage from GPS freezes—saving thousands of fuel liters per commercial fleet annually.
Sovereign Positioning (Atmanirbhar Bharat): Fuses seamlessly with India's indigenous NavIC constellation, ensuring domestic navigation independence.
PM Gati Shakti Alignment: Provides uninterrupted tracking along national multimodal freight tunnels (Atal Tunnel, Zojila Tunnel, DFC).

EMPIRICAL PROOF & QUANTIFIABLE IMPACT	93.8% - Pass Rate(60/64 held-out sequences) (IO-VNBD Dataset)	0.02% - Min Tunnel Drift(IO-VNBD)	< 5.8% — Battery / Hour	₹0 — Hardware Cost	0.50 m/s — Speed MAE

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RESEARCH AND REFERENCES

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Reference	Key Contribution	Relevance to IDR
Titterton & Weston (2004)	Inertial navigation & Kalman filtering	Foundation for **IMU + EKF fusion**
Bai et al. (2018)	Temporal Convolutional Networks	Supports **IMU-based motion & speed**
Groves (2013)	GNSS–INS multisensor integration	Enables **positioning during GNSS loss**
Newson & Krumm (2009)	Robust map matching	Supports **road-based trajectory correction**
Android Developers (2022)	Smartphone sensor characteristics	Guides **IMU & barometer data acquisition**
Onyekpe et al. (2021)	Inertial Odometry Vehicle Navigation Benchmark Dataset (Elsevier)	Primary benchmark validating our 93.8% pass rate (<10% drift threshold)

Production APK & Drive

Direct access to compiled binaries and quantized neural weights:
Flutter App: IDR.apk (60 FPS HUD)
Native Core: C++ Math Engine (200 Hz)
ONNX Model: 2.1 MB INT8 quantized weights

APP **MODEL**

GitHub Source Repository

Complete source code, training scripts, and automated test suite:
Core Stack: Dart EKF + C++20 SIMD Engine
Test Suite: 136 Automated Unit & E2E Tests

GITHUB

Demo & Presentation Video

Video

Training Corpora & Field Data

Multi-dataset curriculum & active Indian road testing:
Public: IO-VNBD (241 trips), DriverSVT (9.1 GB)
OBD-II Truth: UAH-DriveSet synchronized logs
In-House: Smartphone MEMS Telemetry on Indian roads